

Dr. VICTOR HUGO ALMANZA VELOZ

Position: Research Associate. Instituto de Ciencias de la Atmósfera y Cambio Climático
Universidad Nacional Autónoma de México
Circuito Exterior s/n, 04510
SNII 1, Pride B
victor.almanza@atmosfera.unam.mx
Tel: +52 (55) 5622 4800, Ext. 81825

RESEARCH INTEREST

I am interested on atmospheric chemistry, air quality modeling, and data assimilation to develop operational early-warning systems aimed at responding to extreme pollution events and supporting public health decision-making. I particularly investigate the formation and transport of secondary air pollutants (focusing on SOA and PAN) through the integration of atmospheric composition observations, chemical transport modeling and advanced data assimilation techniques with particular emphasis on processes occurring in the middle troposphere.

The aim is to improve the representation of atmospheric processes, strengthen air quality forecasting and response, and provide actionable scientific evidence for air quality management and public health decision-making in Mexico.

EDUCATION

Sc. D. Atmospheric Science Instituto Mexicano del Petróleo (IMP), Mexico - Dissertation title: "Tula Refinery sour gas flaring emissions: numerical estimation and regional air quality modeling" - Authored 3 publications in peer-reviewed journals	2014
M.S. Atmospheric Science IMP, Mexico	2006
B.S. Environmental Engineering Instituto Politécnico Nacional (IPN), Mexico Class Rank: 1	2002

PROFESSIONAL EXPERIENCE

<u>Current position</u> Instituto de Ciencias de la Atmósfera y Cambio Climático (ICAYCC), UNAM <i>Research Associate</i>	2020 - current
Centro de Investigación Científica y de Educación Superior de Ensenada <i>Consultant</i>	2019 - 2020
LTM Center for Energy and the Environment <i>Research Scientist</i>	2018 - 2020
Massachusetts Institute of Technology Molina Center for Energy and the Environment <i>MIT Molina Postdoctoral Fellow</i>	2013 - 2018

PROJECTS

ICAYCC UNAM (Pi/Co-PI)

2020 – current

Data Assimilation (CAME) ongoing

- Co-Principal investigator. Developing an operational air quality forecasting system based on the 3D variational approach for Central Mexico. Funded by the Environmental Commission of the Megalopolis (CAME).

Chemical Reanalysis (PAPIIT UNAM) ongoing

- Principal investigator. Analyzing peroxyacetyl nitrate (PAN) long-term data from chemical reanalysis for the Megalopolis of Central Mexico.

Data Assimilation (PAPIIT UNAM) completed

- Principal investigator. Joint Project in collaboration with NASA Ames and University of Colorado Boulder for conducting ensemble Kalman filter data assimilation using WRF-Chem/DART.
- Conducting ensemble model evaluation using UPP/MET.

Data Assimilation (DGTIC-UNAM)

- Principal investigator. Organized the research visit of Dr. Arthur Mizzi to ICAYCC in November 2022.
- Implemented the forward operator of MaxDOAS NO₂ observations into WRF-Chem/DART in collaboration with Dr. Mizzi.
- Part of the results presented at the *10th International Workshop on Air Quality Forecasting Research 2021* by Dr. Mizzi.

Impact of COVID-19 emissions reductions in the Mexico Megalopolis (DGTIC-UNAM)

- Principal Investigator. Key findings included the offset of massive emissions reductions and a decrease in PBL height by regional biomass burning emissions.
- Some results were presented in virtual workshops organized by government officials of the Environmental Secretariat of the Mexico Megalópolis.

LTM Center for Energy and the Environment

2019 - 2020

UCMEXUS-CONACYT Project

- Principal Co-Investigator together with Dr. Lynn Russell from UC San Diego, in organizing and participating in a capacity building workshop and an international symposium of air quality at UNAM Mexico.
- Responsible of designing the content of the workshop in collaboration with Ravan Ahmadov (NOAA), Douglas Loewe (U. of Manchester) and Arthur Mizzi (U. of Colorado)
- Co-participant of the capacity building workshop for using the WRF/WRF-Chem model in collaboration with Ravan Ahmadov (NOAA), Douglas Loewe (U. of Manchester) and Arthur Mizzi (U. of Colorado).

CICESE

2019 -2020

- Responsible of evaluating the WRF/WRF-Chem model performance of the operational chemical forecast for the Gulf of Mexico using MET/UPP.
- Lead the evaluation of the WRF-FLEXPART model using Chaman cluster at CICESE.

MIT / MCE2

2014 - 2018

NSF Black Carbon Project

- Led the project team in downscaling the NASA GISS-E2-R model to conduct regional climate simulations with WRF for biomass burning applications. Wrote the computer scripts to set-up, run, post-process, evaluate and analyze the data. These results were used to estimate future biomass burning emissions.
- Led the modeling team to evaluate the WRF-Chem model performance using annual data of meteorology and air quality from monitoring networks, remote sensing data from the Tropical Rainfall Measuring Mission (TRMM) and cloud fraction from MODIS products

CARB CalEnviroScreen Project

- Collaborated with a multi-disciplinary team in designing, writing, discussing and reviewing the trajectories analyses of a research proposal funded by the California Air Resources Board (CARB). Led the modeling activities and developed all the computer scripts to efficiently run FLEXPART-WRF for the US-Mexico Border region. Identified potential emission sources with direct benefit to the CalEnviroScreen tool.

GEF SLCF Project

- Led the model evaluation with MET-UPP for the project on SLCPs sponsored by the Global Environment Facility (GEF). These activities were crucial for the modeling set-up. Analyzed the results and wrote a section for the final report.
- Automated the WRF model evaluation using the NCAR's Model Evaluation Tools (MET) package and UPP in the Yellowstone cluster and upgraded to Cheyenne cluster (NCAR). Wrote the computer scripts to unify data formats, set-up, run, post-process and visualize data and results at all stages of the evaluation process.

CAME Ozone Modeling Project

- Collaborated with the modeling team to modify the code of WRF-Chem for analyzing the influence of vertical diffusion on model performance of chemical fields.

MCE2-EDF Methane Emissions Inventory

- Estimated methane emissions for the oil and gas industry in Mexico in collaboration with Mexican Ministry of the Environment and the Environmental Defense Fund (EDF). The QA/QC protocol improved the estimates and location of emission sources with a direct benefit for the Mexican National Emissions Inventory and global emissions inventories. Analyzed the results and wrote several sections of the final report.

MILAGRO Project

- Led the backward trajectories analysis with FLEXPART-WRF using data from the Cal-Mex and MILAGRO field campaign. Wrote the computer scripts to set-up, run, and post-process the results. Improved the interpretation of chemical data and source attribution in the US-Mexico border and the outskirts of MCMA
- Estimated soot emissions from Refinery flares at industrial scale with CFD and conducted the reactive modeling of emissions with WRF-Chem.

ECAIM Project

- Collaborated with scientists from Mexico and Saint Louis University in reviewing some of the main findings from the MILAGRO and MCMA-2003 field campaigns as part of a research project in Mexico. Wrote the final report and highlighted recommendations for modeling and deployment of measuring equipment.

MARPOL Project

- Actively contributed to the QC and modeling activities in an inter-disciplinary project for the Commission for Environmental Cooperation (CEC) on the MARPOL agreement. Improved the model-ready files of ship emissions which were the basis for the modeling activities. The results were presented to officials from the Mexican Navy and the Mexican Secretariat of Environment.

IMP WRF-Chem Workshop

- Invited to Lecture at the Mexican Institute of Petroleum (IMP) on using, installing and running WRF-Chem. Prepared all the workshop material, including the model installation and set-up of using the IMP's workstations.

IMP**2006 – 2013*****Graduate Research Assistant***

- Estimated for the first time in Mexico, soot and combustion by-products emissions of a refinery sour gas flare at industrial scale using Computational Fluid Dynamics. Modified the code of the combustion model in OpenFOAM (C++) to include radiation from soot. The results suggested a significant underestimation of VOCs from flaring activities in Mexico and a potential contribution of flaring activities to the regional ozone levels.
- Assimilated observational data from the MILAGRO field campaign (radiosondes, radar wind profilers and surface measurements) and global upper and surface data from NCEP ADP into WRF-Chem. Wrote the scripts to run the Objective Analysis (OBSGRID) in Fortran and bash.
- Designed a sensitivity study to select an optimal configuration for WRF-Chem.

- Modified WRF-Chem for 1-way nesting simulations with NDOW before this feature was included in a later version. This significantly saved CPU time
- Identified a potential emission source in the outskirts of Mexico City using back-trajectory analyses.

IMP projects

2004 – 2006

- Member of team that estimated the refractive index of sub-micrometric particles using inversion methods based on meta-heuristic algorithms (simulated annealing and genetic algorithms). Wrote the computer programs, analyzed and discussed results. Collaborated in preparing the final report. The results significantly improved previous estimates from scientists at IMP.
- Lead a project team to install, test and run MM5 using the Intel Fortran compiler. Wrote the final report. This work set the basis to install WRF.
- Collaborated with a project team to estimate reaction rates of butadiene with the hydroxyl radical using quantum chemistry methods (Gaussian) in CRAY platform. Analyzed and discussed the results.

UPIBI – IPN

1998 – 2002

Undergraduate Research Assistant

- Applied non-linear time series analysis to air pollutants databases in Mexico. Estimated the Lyapunov exponent, correlation dimension and multifractal spectrum. Some of the results suggested similarity to the Zaslavski Map and were published in the Mexican Journal of Physics. Wrote the research paper and analyzed the results.

COMPUTER SKILLS

- Linux, WRF/WRF-Chem, OBSGRID, FLEXPART-WRF, MET, UPP, VAPOR, ParaView, OpenFOAM, TISEAN, Fortran, MATLAB, OCTAVE, NCO, CDO, NCL, wgrib, bash shell, awk, sed, gnuplot, MS Office.

CERTIFICATE COURSES

- Earth System Predictability Across Timescales Workshop, April 10 – 12, 2024, NCAR, Colorado, EU, virtual.
- MUSICA Tutorial (online) 2021 – 2022
- Chimere Tutorial (online) 2021
- Hysplit Tutorial (online) 2021
- MIT Translational Fellows Program, 2018
- Global Warming Science, MITx, 2016.
- Satellite Remote Sensing of Particulate Matter Air Quality: Data, Tools, Methods and Applications (Aka AOD-PM), NASA, 2015.

SCIENTIFIC PUBLICATIONS

Arellano-Hernández, E.J., **Almanza, V.**, Jazcilevich, A. et al.: Use of public health benefits to design air pollution emission abatement strategies. Sci. Rep. 16, 24251, 2026. <https://doi.org/10.1038/s41598-026-55037-3>.

Maldonado-Paccheco, G., **Almanza, V.**, Ruiz-Suárez, L. G., and García-Reynoso.: Assimilating chemical observations in the Mexico City Metropolitan Area using WRFDA-Chem., Geofísica Internacional. 65(3), 2359–2376. 2026., FI = 0.5.

Garcia, A.; **Almanza, V.**, Advances in Operational Air Quality Forecasting for Mexico City.: Integration of Updated Emissions, Temporal Profiles, and Initial Conditions. Atmósfera. 40, 153-165, 2026. FI = 1.1. Q3.

Garcia R, Andraca GL, Cerón JG, Cerón RM, Espinosa Fuentes MdL, Schiavo B, **Almanza-Veloz V**, Barrera-Huertas H, Torres-Jardon R, Mugica-Alvarez V. Assessment of Carbonyl Compound Levels in Indoor Environments of Residential Buildings in Mexico City: Case Study on the Effects on Health and Quality of Life During Remote Work. Sustainability.18(1):270.2026;F.I=2.5,Q2. <https://doi.org/10.3390/su18010270>

Taquet, N., Stremme, W., González del Castillo, M. E., **Almanza, V.**, Bezanilla, A., Laurent, O., Alberti, C., Hase, F., Ramonet, M., Lauvaux, T., Che, K., and Grutter, M.: CO₂ and CO temporal variability over Mexico City from ground-based total column and surface measurements, *Atmos. Chem. Phys.*, 24, 11823–11848, 2024. F.I = 5.2, Q1., <https://doi.org/10.5194/acp-24-11823-2024>

Almanza, V., Ruiz-Suárez, G., Torres-Jardón, R., García-Reynoso, A., y Hernández-Paniagua, I. Y.: Influence of biomass burning on ozone levels in the Megalopolis of Central Mexico during the COVID-19 lockdown. *Journal of Environmental Sciences.*, 143, 99-115, 2024. F.I = 5.9 , Q1. <https://doi.org/10.1016/j.jes.2023.07.031>

Garcia, A.; **Almanza, V.**, Tejeda, D. and Alvarado-Castillo, M.: Impact of the No-Driving Day Program on Air Quality in a High-Altitude Tropical City: The Case of the Toluca Valley Metropolitan Area. *Atmosphere* 15, 437., 2024. F.I = 2.5, Q2. <https://doi.org/10.3390/atmos15040437>

García-Reynoso A., **Almanza, V.** A new approach to estimate air moisture, *Educacion Química*, accepted for publication, 2023

Zavala-Araiza, D., Omara, M., Gautam, R., Smith, M., Pandey, S., Aben, I., **Almanza-Veloz, V.**, Conley, S., Houweling, S., Kort, E., Maasakkers, J., Molina, L., Pusuluri, A., Scarpelli, T., Schwietzke, S., Shen, L., Zavala, M., and Hamburg, S.: A tale of two regions: Methane emissions from oil and gas production in offshore/onshore Mexico, *Environmental Research Letters*, **16**, 024019, 2021.

Hernandez-Paniagua, I., Valdez, I., **Almanza, V.**, Rivera, C., Grutter, M., Stremme, W., García, A., and Ruiz-Suarez, L.G., Impact of the COVID-19 Lockdown on Air Quality and Resulting Public Health Benefits in the Mexico City Metropolitan Area. *Frontiers in Public Health*, 9:642630, 2021.

Aguilar-Dodier, L. C., Castillo, J.E., Quintana P.J., Montoya, L., Molina, L., Zavala, M., **Almanza-Veloz, V.**, and Rodríguez-Ventura, J.G.: Spatial and temporal evaluation of H₂S, SO₂ and NH₃ concentrations near Cerro Prieto geothermal power plant in Mexico, *Atmospheric Pollution Research*, 11, 1, 94-104, 2019.

Salcedo D., Castro, T., Bernal, J. P., **Almanza-Veloz, V.**, Zavala, M., González-Castillo, E., Saavedra, M. I., Perez-Arvizu, O., Díaz-Trujillo, G. C., and Molina, L. T.: Using trace element content and lead isotopic composition to assess sources of PM in Tijuana, Mexico, *Atmos. Environ.*, 132, 171 – 178, 2016.

Almanza, V. H., Molina, L. T., Li, G., Fast, J., and Sosa, G.: Impact of external industrial sources on the regional and local SO₂ and O₃ levels of the Mexico megacity, *Atmos. Chem. Phys.*, 14, 8483-8499, doi:10.5194/acp-14-8483-2014, 2014.

Almanza V. H., Battyshin I., and Sosa, G: Multi-criteria selection of an air quality model configuration based on quantitative and linguistic evaluations. *Expert Systems with Applications*, 41, 869-876, 2014.

Almanza, V. H., Molina, L. T., and Sosa, G.: Soot and SO₂ contribution to the supersites in the MILAGRO campaign from elevated flares in the Tula Refinery, *Atmos. Chem. Phys.*, 12, 10583-10599, doi:10.5194/acp-12-10583-2012, 2012.

Almanza-Veloz, V. H., and Cabrera-Llanos, A. I.: Reconstrucción del atractor para un contaminante primario de la zona metropolitana de la ciudad de México, *Rev. Mex. Fis.*, 51(2), 22-31, 2005.

TECHNICAL REPORTS

DGTIC – UNAM: Almanza V.: Impacto del periodo de confinamiento por COVID-19 en precursores de ozono, partículas finas y aerosoles orgánicos secundarios, Proyecto LANCAD-UNAM-DGTIC-395, Prepared for DGTIC UNAM, 2022.

CICESE - UNAM: Almanza V.: Versión Funcional del Software Para Realizar Pronósticos Meteorológicos de Calidad del Aire: Evaluación Química. Project CIGOM, prepared for CICESE, 2020.

CICESE - UNAM: Almanza V.: Versión Funcional del Software Para Realizar Pronósticos Meteorológicos de Calidad del Aire: Evaluación Meteorológica. Proyecto CIGOM, preparado para el Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE), 2020.

CARB: Quintana, J., Nara, A., Russell, L., Zavala, M., Almanza, V., Molina, L., Quintana, J.: Improving the CalEnviroScreen Score at the US-Mexico Border, prepared for California Air Resources Board, 2019.

MCE2-INECC (Molina Center for Energy and the Environment and Instituto Nacional de Ecología y Cambio Climático): Molina, L., Zavala, M., Lei, W., Almanza, V.: Integrated responses to short lived climate forcers promoting clean energy and energy efficiency., Prepared for United Nations Environment Programme (UNEP). 2016 (updated 2018).

Comission for Environmental Cooperation (CEC). Molina, L., Zavala, M., Lei, W., Almanza, V.: Reducing Emissions from Goods Movement via Maritime Transportation in North America: Evaluation of the Impacts of Ship Emissions over Mexico, Montreal, Canada, 2018.

MCE2 (Molina Center for Energy and the Environment): Molina, L., Zavala, M., Lei, W., Almanza, V.: Characterization of oil and gas methane emissions sources in Mexico. Prepared for Mexican Ministry of Environment and Environmental Defense Fund, 2017.

LTMCE2 (LTM Center for Energy and the Environment): Molina, L., Zavala, M., Lei, W., Almanza, V.: Evaluación de los impactos en la concentración de ozono por la aplicación de estrategias integradas de control de emisiones en la Megalópolis. Final Report INECC/LPN-009/2017. Mexico, 2017.

MCE2 (Molina Center for Energy and the Environment): Molina, L., Zavala, M., Lei, W., Almanza, V.: Beneficios en el clima y la salud por la mitigación de Contaminantes Climáticos de Vida Corta (CCVC): El caso del carbono negro del sector transporte en México. Preparado para LARCI, 2016.

Instituto Nacional de Ecología y Cambio Climático (INECC): Molina, L., Zavala, M., Lei, W., de Foy, B., Almanza, V.: Estudios de Calidad del Aire y su Impacto en la Región Centro de México (ECAIM). Informe Final, Tomo II. Centro de Ciencias de la Atmósfera, UNAM, 2015.

SCIENTIFIC PRESENTATIONS

- **Virtual workshop: Meteorological factors conducive to air quality episodes in the Megalopolis (oral):** Meteorological and chemical data assimilation in forecast models. May 2023. Organized by the Environmental Secretariat and the Megalopolis Secretariat (SEMARNAT and CAME)
- **Research activities in atmospheric sciences.** March 2023. Tecnológico de Monterrey CDMX.
- **Joint Science Meeting for TEMPO, GeoXO ACX, & TOLNet (poster):** Grutter, M., Stremme, W., Bezanilla, A., Ambriz, R., Taquet, N., Ojeda, Z., Arellano, J., García, A., Pavia, R., Hernández, L., **Almanza, V.**, Amador, O., Salazar, A.: Validation Capacities in Mexico, Alabama, EU. Mayo 1 – 5, 2023.
- **Virtual seminar: Satellite monitoring for air quality management (oral):** Satellite data applications in air quality modeling, data assimilation and emissions estimation: challenges and opportunities. October 2022. Organized by the Megalopolis Secretariat (CAME)
- **1st Student Congress ICAYCC (poster):** Hernández, L., Bezanilla, A., Taquet, N., Stremme W., Gonzalez del Castillo, E., Grutter, M., **Almanza, V.**, García-Zuber, A., Campos-Pineda, M.: Methane emissions estimation in Bordo de Xochiaca, October 2022.
- **Virtual workshop, Current knowledge of scientific basis for air quality management in the Megalópolis (oral):** Regional impact of emissions reductions in ozone and PM2.5 in the Megalopolis during the lockdown April 2022.
- **20th Annual CMAS Conference 2021 (poster):** **Almanza, V.**, and Garcia-Reynoso, A.: The COVID-19 lockdown period in the Mexico Megalopolis: impact on surface SOA and Ozone, November 2021.
- **10th International Workshop on Air Quality Forecasting Research 2021 (oral):** Garcia-Reynoso, A., **Almanza, V.**, Grutter, M., Maldonado, G., Mar-Morales, B., and Rodríguez, A.: Air Quality Evaluation System: Central México Case Study, October 2021.
- **10th International Workshop on Air Quality Forecasting Research 2021 (poster):** Turrent, C., Garcia, A., Domínguez, A. and **Almanza, V.**: Dispersion of Atmospheric Pollution from Surface Oil Burns in the Gulf of Mexico, October 2021.

- **16th IGAC Scientific Conference 2021** (poster): **Almanza, V.**, and Garcia, A.: Impact of the COVID-19 lockdown period in surface Ozone, PM_{2.5}, and SOA in the Mexico Megalopolis, Septiembre 2021.
- **American Geophysical Union (AGU) 2020 Annual meeting** (poster): Zavala-Araiza, D., Omara, M., Gautam, R., Smith, M., Pandey, S., Aben, I., **Almanza-Veloz, V.**, Conley, S., Houweling, S., Kort, E., Maasakkers, J., Molina, L., Pusuluri, A., Scarpelli, T., Schwietzke, S., Shen, L., Zavala, M., and Hamburg, S: Empirical assessment of methane emissions from oil and gas infrastructure in Mexico, December 2020, USA.
- **American Geophysical Union (AGU) 2017 Annual meeting** (poster): **Almanza, V.**, Zavala, M., Lei, W., Shindell, D., and Molina, L.T.: Evaluation of WRF performance driven by GISS-E2-R global model for the 2014 rainy season in Mexico, AGU Fall meeting, Dec 2017, New Orleans, USA.
- **European Geophysical Union (EGU) Annual Meeting 2016** (poster): Zavala, M., **Almanza, V.**, Garcia, A., Jazcilevich, A., Lei, W., and Molina, L. T.: Impacts of black carbon and co-pollutant emissions from transportation sector in Mexico City, EGU General Assembly, April 18th 2016, Vienna, Austria.
- **Mitigation of SLCPs in Mexico 2016** (attendee): **Almanza, V.**: Benefits on Climate and Health by Mitigation of Short-Lived Climate Pollutants: The Case of Black Carbon of the Transportation Sector in Mexico. Cd. De Mexico, April 2016.
- **Hidalgo working meeting 2016 (oral):** **Almanza, V.**, “Main findings during MILAGRO campaign in Hidalgo: Part 2”, First Meeting of Studies and Projects on air pollution in the Tula-Tepeji Area (Pachuca) Mexico. Parque Ecológico Cubitos, Mexico, February 2016.
- **GEF Final Project Meeting 2015** (oral): **Almanza, V.**: “Modeling of SLCPs emission scenarios”, Integrated responses to short-lived climate forcers (SLCFs) promoting clean energy and energy efficiency in Mexico. Torre de Ingeniería UNAM, Mexico, November 2015.
- **US EPA International conference 2015** (attendee): United States Environmental Protection Agency's 2015 International Emission Inventory Conference, Air Quality Challenges: Tackling the Changing Face of Emissions”. Training sessions: Nonpoint source emissions inventory tools and the nonpoint oil and gas emission tool, April 2015.
- **ECAIM project meeting 2015** (oral): **Almanza, V.**: “What can we learn from MILAGRO?”, Second expert meeting on Air Quality and the impacts on the Crown Region of Central Mexico, April 23 2015, Mexico City, Mexico.
- **ECAIM project meeting 2014** (oral): **Almanza, V.**: “Evaluation of the air quality modeling”, First expert meeting on Air Quality and the impacts on the Crown Region of Central Mexico, November 2014, Mexico City, Mexico.
- **EGU 2014** (poster): **Almanza, V.**, Molina, L.T., Li, G., Fast, J., and Sosa G: “Impact of external industrial sources on the regional and local air quality of Mexico Megacity in terms of SO₂ and O₃”. EGU General Assembly 2014, April 2014, Vienna, Austria.
- **GEF project meeting 2014** (oral): **Almanza, V.**: “Modeling emission reduction scenarios”, Third Project Meeting: Integrated responses to short-lived climate forcers (SLCFs) promoting clean energy and energy efficiency in Mexico. Torre de Ingeniería UNAM, Mexico, July 2014.
- **EGU 2013** (oral): **Almanza, V.**, Molina, L.T., Li, G., Fast, J., and Sosa G: “Contribution of Tula Refinery flaring emissions to the Mexico megacity”. April 2013. Vienna, Austria.
- **62nd Annual Meeting APS 2009** (oral): **Almanza, V.** and Sosa, G. “Estimating soot emissions from an elevated flare”, November 24th, 2009, MN, USA.
- **MILAGRO meetings 2008/2009** (oral).

AWARDS AND HONORS

- **CONACYT SNI** (Tier 1),
- **PRIDE B** (2020)

- **MIT Molina Fellowship** (2013-2018),
- **CONACYT SNI** (Candidate; 2017),
- **Keith Runcorn Travel Award for Non-Europeans** (KRTA; EGU 2013),
- **Travel grant APS division of Fluid Dynamics** (2009),
- **Best bachelor students in Mexico** award (class rank: 1; 2002),
- **Graduate student scholarship** (IMP-CONACYT; 2004-2012).

SELECTED SUPERVISION – THESIS ADVISEES (COMPLETED AND IN PROGRESS)

Bachelor

Ignacio Vargas Gonzalez. Modelación numérica de ozono y partículas en el área urbana de Puebla de un episodio registrado en 2019 durante la temporada de incendios. Facultad de Química. Ingeniería Química. Graduated September 2025. Director.

Hugo Miranda Cano. Asimilación de datos de monóxido de carbono del sensor TROPOMI en la Megalópolis. Facultad de Ciencias. Licenciatura en Matemáticas. Completed. Director.

Berenice Cano Mendoza. Evaluación de un Índice de Estancamiento en la Zona Metropolitana del Valle de México para episodios de ozono. In review. Facultad de Ciencias. Licenciatura en Ciencias de la Tierra. Director.

Sandro Lescieur López. Mejora del pronóstico de calidad del aire en la Zona Metropolitana del Valle de México usando aprendizaje automático. In progress. Facultad de Ciencias. Licenciatura en Matemáticas. Director.

Sinuhe Alex Quintero Carbajal. Clasificación de días de estancamiento en la ZMVM usando Support Vector Machines (SVM). In progress. Facultad de Ciencias. Licenciatura en Matemáticas Aplicadas. Director.

Diana Laura Mendoza Jiménez. Reducción del sesgo del pronóstico de calidad del aire en la Zona Metropolitana del Valle de México usando Random Forest. In progress. Instituto Politécnico Nacional (ESIA Ticomán). Ingeniería Geofísica.

Esmeralda Ruby Medina Morales. Evaluación del impacto de la dispersión de ceniza del volcán Popocatepetl para el periodo 2023 – 2024 en la Ciudad de México. Facultad de Química. Ingeniería Química. Supervisor Técnico.

Ph. D

José Heriberto Vélez Garza. Estudio regional de la interacción del transporte de partículas contaminantes PM10 y PM2.5 mediante el modelo WRF-Chem en la actividad fotosintética en *Cordia boissieri* en la Zona Metropolitana de Monterrey, Nuevo León, México. En curso, recién inicia. Universidad Autónoma de Nuevo León (UANL). Departamento de Ecología de la Facultad de Ciencias Biológicas-Unidad B.

Note: Co-advised with Dr. José Ignacio González Rojas (UANL).

Master in Science

Miguel Angel González Sánchez. Análisis de Series de Tiempo Ambientales mediante Cadenas de Markov. En proceso de elaboración. Universidad Autónoma Metropolitana Campus Iztapalapa (UAM-I). Maestría en Ciencias (Matemáticas Aplicadas e Industriales)

Nota: Co-advised with Dr. José Héctor Morales Bárcenas (UAM-I)

Posdoctorate

Dr. Gilberto Maldonado Pacheco. Asimilación de datos de CO y NO2 usando WRFDA-Chem. Proyecto CAME: Ensemble para alerta temprana de episodios y pronóstico de calidad del aire. Componente 1. “Asimilación de datos de química atmosférica y meteorológicos para la región de la Megalópolis”. (09/2024 – 05/2026).

LEADERSHIP AND PROFESSIONAL ACTIVITIES

Postdoctoral Fellow

2013 -2018

- Worked independently on a research proposal aimed at modifying the chemistry solution in a chemistry transport model. Researchers from PNNL, University of Iowa, and NOAA evaluated the technical merit.
- Lectured at the Mexican Institute of Petroleum (IMP).
- **Reviewer for Energy & Fuels** (2016), **Urban and Climate** (2017) and **Atmospheric Pollution Research** (2017).

Graduate Research

2010

- Led the compilation, installation, testing and running of Linux (several distributions), MM5, WRF/WRF-Chem, MPICH, Intel Fortran, OpenFOAM, SALOME, and FLEXPART-WRF.

GURME-WMO workshop (Mexico)

2009

- Led the installation and testing of Linux and WRF-Chem in 33 PCs for the workshop at the Secretariat of Environment in Mexico (SEMARNAT). Worked in collaboration with Dave Gill (NCAR) and Steven Peckham (NOAA).

Undergraduate research

2002 - 2004

- New generations of students continued my research on non-linear time series analysis of atmospheric pollutants